
ME 355: Internal Combustion Engine

Course Code	ME 355
Course Name	Internal Combustion Engine
L-T-P-C	3-0-0-3
Prerequisites	Nil
One-time approval	8th Senate (Oct 2015)

Course Contents

- **Introduction** : Classification of I.C. engine, Fundamental difference between S.I. and C.I. Engines, Comparison of two stroke and four stroke engines, various components and their functions, Types of efficiency, indicated and brake power, theory of carburetion, Air Standard cycles (Diesel, Otto, Dual, Stirling, Brayton) and their comparison, measurement and testing techniques. Thermodynamics of fuel-air cycles, real cycles, various losses in actual engines. Measurement of Indicated power, brake power, fuel consumption, energy balance analysis. (12 Lectures)
- **Combustion and control**: Combustion processes in SI engine and its various stages, spark ignition, normal and abnormal combustion, knock pre-ignition, combustion stages in CI engines, ignition delay, types of combustion systems, Fuel spray behaviour, Exhaust emissions, its measurement and control. (8 Lectures)
- **Engine cooling and lubrication system**: Temperature distributions of various components, heat transfer theory, parameters affecting engine heat transfer, need and type of cooling systems. Engine Lubrication System: Frictional losses, various types of lubrication systems, lubrication of two stroke engines (4 Lectures)
- **Fuel Injection and Ignition systems**: Requirement of Air fuel mixing, fuel properties and its measurement, fuel injection system in SI engines, fuel injection system in CI engines, turbocharging, ignition systems. (8 Lectures)
- **Advances in Road Transport**: Hybrid vehicles; electric vehicles; fuel cell powered vehicles; hydrogen fuelled engines; new combustion Modes for IC engines such as low temperature combustion, homogeneous charge compression ignition, reactivity controlled combustion etc.

Text Books

1. Ganesan, **Internal Combustion Engines**, 4th Edition, Tata McGraw Hill Publishing Co.ltd., 2012.
2. John B. Heywood, **Internal Combustion Engine Fundamental**, Tata McGraw-Hill Education, 2011.

References

1. M L Sharma and M P Mathur, **Internal Combustion Engines**, Dhanpat Rai Publications, 2016.
2. Richard Stone, **Introduction to Internal Combustion Engines**, Society of Automotive Engineers, 1999.
3. Paul W. Gill and James H. Smith, **Fundamental of Internal Combustion Engines**, 4th Edition, Oxford & IBH Publishing Company Pvt. Limited, 2007.